Homework #3 – Simple DL Workflow in Kubeflow

- Objective:
  - Develop Container and Kubernetes artifacts perform DL training and DL inference hosting in GKE: Google Kubernetes Engine.
  - The work demands self-driven deep dive into Kubernetes, Kubeflow concepts and components.
- Submission:
  - Dockerfile, K8S yaml files, instructions on how to build and run the service on K8S clusters on GCP-GKE. Screen captures on how the app run.
  - Document your work and report on your experiences: e.g.
    - What Kuberentes controllers did you use for training and inference and why?

Homework #3 – Architecture Aspect

- Application containers
  - Training: training program, container definition file (dockerfile), training job yaml file
  - Inference: inference program, dockerfile, service yaml file, deployment yaml file
    - Training and inference program need to store and load the trained model persistently.
    - Host a URL to allow interaction with User.
- User interaction
  - Give any input to the inference engine (a number, a space bar, image file, image id in the data set) to show that the inference is happening in an interactive/controlled manner.
- Persistent storage (PVC) – if required (optional)
  - After creating your cluster, you can create a PVC resource:
    - GCP → Kubernetes Engine → Storage → Persistent storage volume